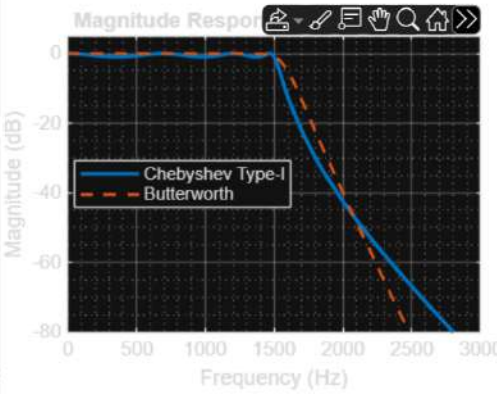
```
filter_mode = 'LPF';
if strcmp(filter_mode, 'LPF')
    fs = 10000;
    fp = 1500;
    fst = 2000;
    Rp = 1.0;
    As = 40;
else
    fs = 10000;
    fp = 3000;
    fst = 2000;
    Rp = 0.5;
    As = 45;
end
Wp = fp / (fs/2);
Wst = fst / (fs/2);
[N_cheby, Wp_cheby] = cheb1ord(Wp, Wst, Rp, As);
[N_butter, Wn_butter] = buttord(Wp, Wst, Rp, As);
if strcmp(filter_mode, 'LPF')
    [b_cheby, a_cheby] = cheby1(N_cheby, Rp, Wp_cheby, 'low');
    [b_butter, a_butter] = butter(N_butter, Wn_butter, 'low');
else
    [b_cheby, a_cheby] = cheby1(N_cheby, Rp, Wp_cheby, 'high');
    [b_butter, a_butter] = butter(N_butter, Wn_butter, 'high');
end
fprintf('=====================================================\n');
fprintf('        DESIGN ANALYSIS RESULTS FOR %s FILTER          \n', filter_mode);
fprintf('=====================================================\n');
```



```
fprintf('Chebyshev Type-I Filter Order (N): %d\n', N_cheby);
fprintf('Butterworth Filter Order (N)   : %d\n\n', N_butter);
disp('--- Chebyshev Type-I Transfer Function Coefficients ---');
disp('Numerator (b):'); disp(b_cheby);
disp('Denominator (a):'); disp(a_cheby);
[H_cheby, f_vec] = freqz(b_cheby, a_cheby, 2048, fs);
[H_butter, ~]   = freqz(b_butter, a_butter, 2048, fs);
mag_cheby  = 20 * log10(abs(H_cheby));
mag_butter = 20 * log10(abs(H_butter));
phase_cheby = unwrap(angle(H_cheby)) * (180/pi);
figure('Color', [1 1 1]);
subplot(2,1,1);
plot(f_vec, mag_cheby, 'LineWidth', 2, 'Color', [0 0.4470 0.7410]); hold on;
plot(f_vec, mag_butter, '--', 'LineWidth', 1.5, 'Color', [0.8500 0.3250 0.0980]);
grid on; grid minor;
title(['Magnitude Response Comparison: ', filter_mode]);
xlabel('Frequency (Hz)'); ylabel('Magnitude (dB)');
ylim([-80 5]);
legend('Chebyshev Type-I', 'Butterworth', 'Location', 'best');
xline(fp, '-r', ['Passband: ', num2 old(fp), 'Hz']);
xline(fst, '-k', ['Stopband: ', num2str(fst), 'Hz']);
subplot(2,1,2);
plot(f_vec, phase_cheby, 'LineWidth', 2, 'Color', [0.4660 0.6740 0.1880]);
grid on; grid minor;
title(['Phase Response: Chebyshev Type-I ', filter_mode]);
xlabel('Frequency (Hz)'); ylabel('Phase (Degrees)');

```

